# General Market Impact Classifier

Solution Overview & Results

## 1 Problem

Given a political tweet and its timestamp, the system must answer two questions:

- Is this tweet market-relevant?
- If so, in which direction does it move each of seven financial assets (gold, equities, BTC, crude oil, wheat, EUR/USD, 2Y Treasury) at the 1m, 5m, and 10m horizons?

## 2 Approach

A single tweet carries several independent signals: who is mentioned, what sentiment it conveys, what kind of event it describes, when it was posted, and how that event propagates through connected markets. Rather than fine-tuning one large model on raw text, the pipeline decomposes these signals into five structured feature extraction layers, combines them into a ~130-feature vector fed to a LightGBM statistical ensemble, and then passes all evidence to a final GPT-4o reasoning layer that synthesises the predictions with live market context retrieved from Tavily.

## 3 Seven-Layer Pipeline

**Layer 1 — Named Entity Recognition (GLiNER)**

Extracts persons, countries, commodities, organizations, and policy terms using zero-shot NER. Each entity is mapped to one or more target asset classes via a curated lookup table. Falls back to keyword matching when GPU is unavailable. Produces 14 features.

**Layer 2 — Financial Sentiment (FinBERT)**

FinBERT is pre-trained on financial corpora and outputs positive/negative/neutral probabilities plus a compound score in $[-1, 1]$. More domain-calibrated than general-purpose sentiment models. Produces 8 features.

**Layer 3 — Event Detection (DistilBERT + Trainable Head)**

A frozen DistilBERT encoder feeds a lightweight two-layer MLP head trained on pseudo-labeled tweets. Classifies 13 event types (trade war, sanctions, monetary policy, geopolitical conflict, election, etc.). A hybrid overlay of 100+ regex rule patterns boosts recall on rare event types. Produces 27 features.

**Layer 4 — Context Enrichment**

Extracts temporal signals (trading session, hour-of-day, day-of-week), tweet stylistic signals (caps ratio, urgency, URL presence), and thread position features. Fetches live macro news via Tavily Search API and extracts thematic flags (tariffs, conflict, rate policy). Produces 18 features.

**Layer 5 — Entity-Event-Asset Graph Reasoning**

Builds a directed weighted graph per tweet using 97 domain-knowledge transmission rules. Signal propagates from entity nodes through event nodes to asset nodes. Sentiment compound score modulates each asset's net signal by ±30%. Produces 21 features.

The ~130 assembled features are passed to a LightGBM ensemble consisting of:

- 1 binary relevance classifier — market-relevant vs. not relevant
- 21 multiclass direction classifiers — 7 assets × 3 timeframes, each predicting −1 / 0 / +1

**Layer 6 — LightGBM Statistical Ensemble**

Trained with 5-fold Stratified K-Fold cross-validation on 2,669 tweets. Class weights compensate for the skewed {−1, 0, +1} label distribution. Feature importances generate per-prediction reasoning strings. All 22 models are independently serialised.

**Layer 7 — LLM Reasoning Synthesis (Azure OpenAI GPT-4o)**

The mandatory final synthesis layer. Receives structured outputs from all six preceding layers and per-asset LightGBM predictions. Fetches real-time market news via Tavily using a timestamp-aware query, assembles a structured prompt, and calls GPT-4o (temperature 0.1, response_format: json_object) to produce a final per-asset direction, confidence score, and natural-language reasoning. The LLM also explicitly classifies market relevance as a first-class output, grounded in live context.

## 4 Key Design Decisions

**Modular layers over end-to-end fine-tuning**

With ~2,700 labeled tweets, fine-tuning a large transformer risks overfitting. Decomposing into interpretable layers lets each component specialise while keeping the final learner small and fast.

**Graph transmission rules**

Structured financial domain knowledge is encoded directly as rules (e.g. US–China trade war → equities bearish, gold bullish). This strong prior compensates for limited training data and generalises to event combinations unseen during training.

**Hybrid event detection**

The trained DistilBERT head handles common event types; 100+ regex patterns cover rare categories (military, crypto policy) that appear too infrequently to learn from data alone.

**Temporal train/test split**

The oldest 80% of tweets train the models; the most recent 20% are held out. This prevents leakage and mirrors real deployment conditions.

**LightGBM for statistical prediction**

Fast training, native class-weight balancing, and interpretable feature importances used to generate per-prediction reasoning strings. Gradient-boosted trees outperform deep models on this tabular feature set at this data scale.

**GPT-4o + Tavily as mandatory synthesis layer**

LightGBM cannot reason about novel event combinations not seen in training data. GPT-4o receives all structured evidence plus live Tavily news context and produces calibrated per-asset verdicts with natural-language explanations. Tavily is essential in

Layer 4 (structured macro flags) and Layer 7 (verbatim news grounding for the LLM).

# 5 Results

Evaluated on 667 held-out test tweets via strict temporal split (no test data seen during training). Layer 6 is evaluated on all 21 classifiers across the full test set. Layer 7 is evaluated on the same 667 tweets for direct comparison.

### Layer 6 — Relevance Classifier (LightGBM)

| Classifier | Metric | Score |
|---|---|---|
| Relevance Classifier | Accuracy | 85.01% |
| Relevance Classifier | ROC-AUC | 86.93% |
| Relevance Classifier | Recall | 57.04% |
| Relevance Classifier | F1 (Weighted) | 84.53% |
| Relevance Classifier | F1 (Macro) | 76.25% |
| Relevance Classifier | CV Accuracy (5-fold) | 95.47% |
| Relevance Classifier | CV F1 (5-fold) | 95.54% |

### Layer 6 — Direction Classifiers (LightGBM, 5m horizon)

| Asset | Accuracy | Bal. Accuracy | F1 (Macro) | MCC |
|---|---|---|---|---|
| wheat | 53.97% | 50.30% | 49.19% | 0.293 |
| btc | 46.93% | 34.36% | 33.29% | 0.018 |
| eurodollar | 43.93% | 41.74% | 40.78% | 0.098 |
| gold | 43.63% | 33.29% | 32.88% | -0.015 |
| equities | 42.58% | 34.27% | 33.99% | 0.006 |
| cl | 40.03% | 37.06% | 36.57% | 0.030 |
| treasury_2y | 35.23% | 34.58% | 34.55% | 0.019 |
| **Mean (all assets &** | **44.14%** | **38.08%** | **37.50%** | **0.068** |

### Layer 7 — Relevance Classification (GPT-4o vs LightGBM)

| Metric | LightGBM (Layer 6) | GPT-4o (Layer 7) |
|---|---|---|
| Accuracy | 86.96% | 85.61% |
| Recall | 57.75% | 45.77% |
| F1 (Weighted) | 86.30% | 84.14% |

### Layer 7 — Per-Asset Direction (GPT-4o vs LightGBM, 5m horizon)

| Asset | LGBM Acc | LLM Acc | LGBM F1 | LLM F1 | Agreement |
|---|---|---|---|---|---|
| wheat | 53.37% | 55.47% | 42.33% | 24.28% | 57.72% |
| treasury_2y | 37.33% | 37.48% | 34.96% | 22.10% | 41.38% |
| eurodollar | 45.88% | 13.79% | 40.70% | 9.86% | 15.89% |
| cl | 40.48% | 17.84% | 30.72% | 15.64% | 12.14% |

| | | | | | |
|---|---|---|---|---|---|
| equities | 44.68% | 13.94% | 34.38% | 12.51% | 10.19% |
| gold | 42.58% | 10.79% | 27.63% | 10.46% | 7.05% |
| btc | 47.98% | 4.95% | 33.74% | 3.84% | 6.30% |
| **Mean** | **44.61%** | **22.04%** | **34.92%** | **14.10%** | **21.52%** |

The relevance classifier is the primary production-grade component, correctly identifying market-moving tweets at 85–87% accuracy across both layers. The directional models are directionally informative on assets like wheat (54%) where geopolitical and agricultural trade events create a clear signal. GPT-4o serves as a reasoning and interpretability engine, providing natural-language justifications and live market grounding via Tavily, and achieves accuracy parity with LightGBM on the wheat and treasury_2y assets.